PRODUCTION IMPLEMENTATION BLUEPRINT

AI CONTENT FACTORY MANUAL

Complete Zero-to-One Deployment, Configuration & Operations Manual for Commercial AI Automation

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Deepak Bagada Systems • Commercial AI Digital Assets

Production Release Standard v1.0.0 • Verified Offline & Zero-Leak Tested

AI Content Factory Pipeline — Implementation Manual

Production Release v1.0.0 | Deepak Bagada Systems

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1. System Overview

The **AI Content Factory Pipeline** is an automated multi-agent content generation and publishing engine. It ingests target topics and keywords, constructs comprehensive 1,400+ word engineering-grade technical guides, generates JSON-LD Schema markup, and stages preview drafts in Slack for 1-click publishing to WordPress, Webflow, or Ghost.

Core Architecture Pillars

- **Zero AI Clichés:** Systematically eliminates generic opening hooks, buzzwords, and repetitive filler.
- **Human-in-the-Loop Control:** Directs article previews to your Slack channel before pushing live to your CMS.
- **Self-Hostable:** Includes Docker compose for 1-click self-hosting with zero platform fees.

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2. Prerequisites & Credentials Checklist

Platform	Required Scope	Where to Obtain
n8n or Make.com	Active Workflow Instance	Self-hosted or Cloud
OpenAI / Claude API	Model execution (`gpt-4o`)	platform.openai.com
WordPress / Webflow	Application Password / API Token	CMS Dashboard $\rightarrow$ Users $\rightarrow$ Application Passwords
Slack	Incoming Webhook	api.slack.com/apps

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3. Environment Configuration (`.env`)

Configure the included `.env.example` :

```
OPENAI_API_KEY=your_openai_api_key_here
SLACK_APPROVAL_WEBHOOK_URL=https://hooks.slack.com/services/YOUR/SLACK/WEBHOOK
PORT=5680
```

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4. Step-by-Step Installation & Import

Step 1: Start Container Stack

```
docker compose up -d
```

Access the environment at `http://localhost:5680`.

Step 2: Import Workflow Blueprint

1. In n8n, go to **Workflows $\rightarrow$ Import from File**.
2. Select `workflows/n8n-content-factory.json`.
3. Bind your OpenAI credentials to the **OpenAI GPT-4o Writing Engine** node.
4. Bind your Slack webhook URL in the **Send Slack Preview & Approval** node.

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5. Verification & Sandbox Run

Trigger an article generation job via terminal:

```
curl -X POST "http://localhost:5680/webhook/content-topic-trigger" \
  -H "Content-Type: application/json" \
  -d '{
    "topic": "Zero-Downtime Database Migration Strategies in Distributed Systems",
    "primaryKeyword": "zero downtime database migration",
    "targetAudience": "DevOps Engineers and Cloud Architects",
    "wordCountTarget": 1500
  }'
```

Expected Output

The endpoint returns:

```
{
  "status": "completed",
  "jobId": "job_98432b",
  "slug": "zero-downtime-database-migration",
  "metaTitle": "Zero-Downtime Database Migrations: Dual-Write & Shadow Schema Guide",
  "metaDescription": "Step-by-step technical blueprints for executing zero-downtime database
  "stagedInSlack": true,
  "timestamp": "2026-10-25T11:37:00.000Z"
}
```


A preview card with word count, title, and excerpt will appear in your Slack channel for editorial review.

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6. Troubleshooting & Error Recovery Matrix

Issue	Root Cause	Fix
Timeout during generation	Long generation time on 1,500+ words	Set execution timeout to 60s in container environment.
`429 Rate Limit`	OpenAI TPM limits	The pipeline uses automatic backoffs. Verify Tier 2+ OpenAI billing status.
WordPress 401 error	Invalid App Password	Re-generate Application Password in WordPress User profile.

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